Appendix A. Asymptotics

Bessel functions, Laplace's method and Stirling's formula, the Airy function, the partition function

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Overview

When a function cannot be computed explicitly, one studies its asymptotic behavior; complex analysis is of crucial help. Typically we study, for large s,

$$I(s) = \int_a^b e^{-s\Phi(x)}\,dx. \tag{1}$$

Three guiding principles:

(i) *Deformation of contour.* If Φ is complex-valued, the integrand oscillates rapidly for large s and cancellations mask the behavior. When Φ is holomorphic one may change the contour so that Φ is essentially real on it, and then read off the behavior (illustrated with Bessel functions).

(ii) *Laplace's method.* If Φ is real and $s > 0$, the main contribution comes from near a minimum of Φ, giving an expansion in terms of the quadratic behavior of Φ there (Stirling's formula, the Airy function).

(iii) *Generating functions.* The asymptotics of a sequence F_n can be studied via

$$F_n = \int_\gamma F(e^{2\pi i z})e^{-2\pi i n z}\,dz,$$

γ a segment of unit length in the upper half-plane, a variant of (1); we obtain the asymptotics of the partition function $p(n)$.

1 Bessel functions

Bessel functions arise in problems with rotational symmetry (e.g. the Fourier transform of a radial function on $\mathbb{R}^d$ involves $J_{d/2-1}$). For $\nu > -1/2$,

$$J_\nu(s) = \frac{(s/2)^\nu}{\Gamma(\nu+1/2)\Gamma(1/2)}\int_{-1}^1 e^{isx}(1-x^2)^{\nu-1/2}\,dx. \tag{2}$$

$J_{-1/2}(s) = \lim_{\nu\to -1/2} J_\nu(s) = \sqrt{2/\pi s}\cos s$ and $J_{1/2}(s) = \sqrt{2/\pi s}\sin s$; J_ν is elementary only for half-integral ν. These examples suggest:

Theorem 1.1

$$J_\nu(s) = \sqrt{\frac{2}{\pi s}}\cos\left(s - \frac{\pi\nu}{2} - \frac{\pi}{4}\right) + O(s^{-3/2}) \text{ as } s\to\infty.$$

It suffices to study

$$I(s) = \int_{-1}^1 e^{isx}(1-x^2)^{\nu-1/2}\,dx. \tag{3}$$

Let $f(z) = e^{isz}(1-z^2)^{\nu-1/2}$ in the plane slit along $(-\infty,-1]\cup[1,\infty)$, with the branch positive on $(-1,1)$. For $s>0$ fixed, Cauchy's theorem on the rectangle $\gamma_{\epsilon,R}$ with vertices $-1+\epsilon$, $1-\epsilon$, $1-\epsilon+iR$, $-1+\epsilon+iR$ (letting $\epsilon\to 0$, $R\to\infty$) gives

$$I(s) = -I_-(s) - I_+(s),$$

$I_\pm$ the integrals over the vertical half-lines from ± 1 upward, oriented upward for I_+ and downward for I_- appropriately. On the contour of I_+, $z = 1+iy$:

$$I_+(s) = ie^{is}\int_0^\infty e^{-sy}(1-(1+iy)^2)^{\nu-1/2}\,dy, \tag{4}$$

and similarly for I_-. In (3), e^{isx} oscillates rapidly; in (4), e^{-sy} decreases rapidly except near $y=0$, so the main contribution comes from y near 0:

Proposition 1.2

Suppose a and m are fixed, with $a > 0$ and $m > -1$. Then as $s \to \infty$

$$\int_0^a e^{-sx} x^m \, dx = s^{-m-1}\Gamma(m+1) + O(e^{-cs}), \tag{5}$$

for some positive c.

Proof. $m > -1$ guarantees convergence at 0. Write $\int_0^a = \int_0^\infty - \int_a^\infty$. The first equals $s^{-m-1}\Gamma(m+1)$ (change variables $x \mapsto x/s$). For the second,

$$\int_a^\infty e^{-sx} x^m \, dx = e^{-cs} \int_a^\infty e^{-s(x-c)} x^m \, dx = O(e^{-cs}) \tag{6}$$

whenever $c < a$. ∎

Proof of Theorem 1.1. $(1-(1+iy)^2)^{\nu-1/2} = (-2iy)^{\nu-1/2} + O(y^{\nu+1/2})$ for $0 \le y \le 1$, and $= O(y^{\nu-1/2} + y^{2\nu-1})$ for $y \ge 1$. Proposition 1.2 with $a = 1$, $m = \nu - 1/2$ (and $\nu + 1/2$), together with (6), gives

$$I_+(s) = i(-2i)^{\nu-1/2} e^{is} s^{-\nu-1/2}\Gamma(\nu+1/2) + O(s^{-\nu-3/2}),$$
$$I_-(s) = i(2i)^{\nu-1/2} e^{-is} s^{-\nu-1/2}\Gamma(\nu+1/2) + O(s^{-\nu-3/2})$$

(with the appropriate sign conventions). Since $J_\nu(s) = \frac{(s/2)^\nu}{\Gamma(\nu+1/2)\Gamma(1/2)}[-I_-(s) - I_+(s)]$ and $\Gamma(1/2) = \sqrt{\pi}$, combining the two conjugate terms gives the theorem: $(-2i)^{\nu-1/2}$ and $(2i)^{\nu-1/2}$ contribute the phases $e^{\mp i\pi(\nu-1/2)/2}$, and the sum is $2^{\nu+1/2} s^{-\nu-1/2}\Gamma(\nu+\frac{1}{2})\cos(s - \frac{\pi\nu}{2} - \frac{\pi}{4})$ up to $O(s^{-\nu-3/2})$. ∎

Under restricted circumstances Proposition 1.2 extends to $\mathrm{Re}(s) \ge 0$:

Proposition 1.3

Suppose a and m are fixed, with $a > 0$ and $-1 < m < 0$. Then as $|s| \to \infty$ with $\mathrm{Re}(s) \ge 0$,

$$\int_0^a e^{-sx} x^m \, dx = s^{-m-1}\Gamma(m+1) + O(1/|s|).$$

(Here s^{-m-1} is the branch that is positive for $s > 0$.)

Proof. First, for $\mathrm{Re}(s) \ge 0$, $s \ne 0$, $\int_0^\infty e^{-sx} x^m \, dx = \lim_{N\to\infty} \int_0^N e^{-sx} x^m \, dx$ exists and equals $s^{-m-1}\Gamma(m+1)$. Write $\int_0^N = \int_0^a + \int_a^N$; the first is entire in s. Since $-\frac{1}{s}\frac{d}{dx}(e^{-sx}) = e^{-sx}$, integration by parts gives

$$\int_a^N e^{-sx} x^m \, dx = -\frac{m}{s}\int_a^N e^{-sx} x^{m-1} \, dx - \Big[\frac{e^{-sx}x^m}{s}\Big]_a^N. \tag{7}$$

As $m < 0$, $\int_a^\infty x^{m-1} dx < \infty$, so letting $N \to \infty$, $\int_a^\infty e^{-sx} x^m \, dx$ is analytic in $\mathrm{Re}(s) > 0$ and continuous on $\mathrm{Re}(s) \ge 0$, $s \ne 0$; it equals $s^{-m-1}\Gamma(m+1) - \int_0^a$ for $s > 0$, hence everywhere there. Finally $\int_0^a = \int_0^\infty - \int_a^\infty$, and by (7), $\int_a^\infty e^{-sx} x^m \, dx = O(1/|s|)$. ∎

(Better error terms, or other ranges of m, require smooth cut-offs to mitigate the end-point $x = a$.)

2 Laplace's method; Stirling's formula

Motivation. Integrals $\int e^{-s\Phi(x)}\psi(x)dx$ with large s are dominated by a neighbourhood of the minimum of Φ. Laplace's method makes this precise, and applied to $\Gamma(s)$ it gives Stirling's formula.

Consider $\int_a^b e^{-s\Phi(x)}\psi(x)\,dx$ with real Φ, and Φ, ψ indefinitely differentiable. Assume there is $x_0 \in (a,b)$ with $\Phi'(x_0) = 0$ and $\Phi''(x) > 0$ throughout $[a,b]$.

Proposition 2.1 (Laplace's method)

Under the above assumptions, with $s > 0$ and $s \to \infty$,

$$\int_a^b e^{-s\Phi(x)}\psi(x)\,dx = e^{-s\Phi(x_0)}\Big[\frac{A}{s^{1/2}} + O\Big(\frac{1}{s}\Big)\Big], \qquad A = \frac{(2\pi)^{1/2}\psi(x_0)}{(\Phi''(x_0))^{1/2}}. \tag{8}$$

Proof. Replacing Φ by $\Phi - \Phi(x_0)$, assume $\Phi(x_0) = 0$. Since $\Phi'(x_0) = 0$, $\frac{\Phi(x)}{(x-x_0)^2} = \frac{\Phi''(x_0)}{2}\varphi(x)$ with φ smooth, $\varphi(x) = 1 + O(x - x_0)$. Near x_0 change variables $y = (x - x_0)(\varphi(x))^{1/2}$; then $dx/dy = 1 + O(y)$ and $\psi(x) = \tilde{\psi}(y) = \psi(x_0) + O(y)$. On a small interval $[a', b'] \ni x_0$,

$$\int_{a'}^{b'} e^{-s\Phi(x)}\psi(x)\,dx = \psi(x_0)\int_\alpha^\beta e^{-s\frac{\Phi''(x_0)}{2}y^2}\,dy + O\Big(\int_\alpha^\beta e^{-s\frac{\Phi''(x_0)}{2}y^2}|y|\,dy\Big), \tag{9}$$

$\alpha < 0 < \beta$. With $y^2 = X$, $dy = \frac{1}{2}X^{-1/2}dX$, and (5), the first integral is $s^{-1/2}(\frac{2\pi}{\Phi''(x_0)})^{1/2} + O(e^{-\delta s})$; similarly the second is $O(1/s)$. On $[a, a']$ and $[b', b]$, $\Phi \geq c > 0$, so those integrals decay exponentially. ∎

(8) extends to complex s with $\mathrm{Re}(s) \geq 0$, using integration by parts to handle the oscillations of $e^{-s\Phi}$:

Proposition 2.2

With the same assumptions on Φ and ψ, the relation (8) continues to hold if $|s| \to \infty$ with $\mathrm{Re}(s) \geq 0$.

Proof. Proceed to (9); the first term is handled by Proposition 1.3 with $m = -1/2$. Observation: if $\Phi \geq 0$ and $|\Phi'(x)| \geq c > 0$ on $[a,b]$, then for $\mathrm{Re}(s) \geq 0$,

$$\int_a^b e^{-s\Phi(x)}\psi(x)\,dx = O\Big(\frac{1}{|s|}\Big) \qquad \text{as } |s| \to \infty; \tag{10}$$

indeed the integral is $-\frac{1}{s}\int_a^b \frac{d}{dx}(e^{-s\Phi})\frac{\psi}{\Phi'}\,dx = \frac{1}{s}\int_a^b e^{-s\Phi}\frac{d}{dx}\Big(\frac{\psi}{\Phi'}\Big)dx - \frac{1}{s}\Big[e^{-s\Phi}\frac{\psi}{\Phi'}\Big]_a^b$, and $|e^{-s\Phi}| \leq 1$. This handles $[a, a']$ and $[b', b]$, where $|\Phi'| \geq c > 0$ (as $\Phi'(x_0) = 0$ and $\Phi'' \geq c_1 > 0$). The second term of (9) has the form $\int e^{-s\frac{\Phi''(x_0)}{2}y^2}y\eta(y)\,dy$ with η differentiable; write it as $\frac{-1}{s\Phi''(x_0)}\int \frac{d}{dy}(e^{-s\frac{\Phi''(x_0)}{2}y^2})\eta(y)\,dy$ and integrate by parts to get $O(1/|s|)$. ∎

For purely imaginary $s = it$ this is the **method of stationary phase**; points with $\Phi'(x_0) = 0$ are **critical points**.

For $\delta > 0$ let $S_\delta = \{s : |\arg s| \leq \pi - \delta\}$, and let $\log s$ be the principal branch.

Theorem 2.3 (Stirling's formula)

If $|s| \to \infty$ with $s \in S_\delta$, then

$$\Gamma(s) = e^{s\log s}e^{-s}\frac{\sqrt{2\pi}}{s^{1/2}}\Big(1 + O\Big(\frac{1}{|s|^{1/2}}\Big)\Big). \tag{11}$$

(With more effort the error term improves to $O(1/|s|)$, with a full asymptotic expansion in powers of $1/s$. Note $\Gamma(s) \sim \sqrt{2\pi}s^{s-1/2}e^{-s}$, the usual form.)

Proof. First let $\mathrm{Re}(s) \geq 0$, with uniform error terms off a neighborhood of 0. For $s > 0$,

$$\Gamma(s) = \int_0^\infty e^{-x} x^s \frac{dx}{x} = \int_0^\infty e^{-x+s\log x} \frac{dx}{x} = e^{s\log s} e^{-s} \int_0^\infty e^{-s\Phi(x)} \frac{dx}{x}, \qquad \Phi(x) = x - 1 - \log x,$$

after $x \mapsto sx$; by analytic continuation $\Gamma(s) = e^{s\log s} e^{-s} I(s)$ for $\mathrm{Re}(s) > 0$, $I(s) = \int_0^\infty e^{-s\Phi(x)} \frac{dx}{x}$. It suffices to show

$$I(s) = \frac{\sqrt{2\pi}}{s^{1/2}} + O\Big(\frac{1}{|s|}\Big) \qquad \text{for } \mathrm{Re}(s) \geq 0. \tag{12}$$

$\Phi(1) = \Phi'(1) = 0$, $\Phi''(x) = 1/x^2 > 0$, $\Phi''(1) = 1$; so Φ is convex, positive, with minimum at 1. Apply Proposition 2.2 on $[1/2, 2]$ with $x_0 = 1$, $\psi(x) = 1/x$: this gives (12) there. On $[0, 1/2]$, since $\Phi'(x) = 1 - 1/x$,

$$\int_\epsilon^{1/2} e^{-s\Phi(x)} \frac{dx}{x} = -\frac{1}{s} \int_\epsilon^{1/2} \frac{d}{dx}(e^{-s\Phi(x)}) \frac{dx}{\Phi'(x)x} = -\frac{1}{s}\Big[\frac{e^{-s\Phi(x)}}{x-1}\Big]_\epsilon^{1/2} - \frac{1}{s}\int_\epsilon^{1/2} e^{-s\Phi(x)} \frac{dx}{(x-1)^2}.$$

Since $\Phi(\epsilon) \to +\infty$ as $\epsilon \to 0$ and $|e^{-s\Phi}| \leq 1$, in the limit

$$\int_0^{1/2} e^{-s\Phi} \frac{dx}{x} = \frac{2}{s} e^{-s\Phi(1/2)} - \frac{1}{s}\int_0^{1/2} e^{-s\Phi} \frac{dx}{(x-1)^2} = O(1/|s|).$$

$\int_2^\infty$ is treated analogously, since $\int_2^\infty (x-1)^{-2} dx < \infty$. So (11) holds for $\mathrm{Re}(s) \geq 0$, $|s| \to \infty$.

To pass to $\mathrm{Re}(s) \leq 0$, $s \in S_\delta$: for $\mathrm{Re}(s) \geq 0$, $s = \sigma + it$, $t \neq 0$, $\log(-s) = \log s - i\pi$ if $t > 0$ and $\log s + i\pi$ if $t < 0$. Hence for $G(s) = e^{s\log s} e^{-s}$,

$$G(-s)^{-1} = e^{s\log s} e^{-s} e^{-si\pi} \ (t > 0), \qquad G(-s)^{-1} = e^{s\log s} e^{-s} e^{si\pi} \ (t < 0). \tag{13}$$

Also

$$\Gamma(s)\Gamma(-s) = \frac{-\pi}{s \sin \pi s}, \tag{14}$$

from $\Gamma(s)\Gamma(1-s) = \pi/\sin \pi s$ and $\Gamma(1-s) = -s\Gamma(-s)$ (Theorem 1.4 of Chapter 6, Lemma 1.2 of Chapter 6). Combining (13), (14) and $(1 + O(|s|^{-1/2}))^{-1} = 1 + O(|s|^{-1/2})$ extends (11) to S_δ (in S_δ, $\sin \pi s$ is comparable to $e^{\pm i\pi s}/(\mp 2i)$ for large $|t|$, and near the negative real axis s is excluded). ■

3 The Airy function

The Airy function appeared in optics (intensity of light near a caustic):

$$\mathrm{Ai}(s) = \frac{1}{2\pi} \int_{-\infty}^\infty e^{i(x^3/3 + sx)} \, dx, \qquad s \in \mathbb{R}. \tag{15}$$

The integral converges and is continuous in s: $\frac{1}{i(x^2+s)} \frac{d}{dx} e^{i(x^3/3+sx)} = e^{i(x^3/3+sx)}$, so for $a \geq 2|s|^{1/2}$,

$$\int_a^R e^{i(x^3/3+sx)} \, dx = \int_a^R \frac{1}{i(x^2+s)} \frac{d}{dx} e^{i(x^3/3+sx)} \, dx, \tag{16}$$

and integrating by parts and letting $R \to \infty$ shows uniform convergence for $|s| \leq a^2/4$; similarly on $(-\infty, -a]$. Replacing $\mathbb{R}$ by $L_\delta = \{x + i\delta : x \in \mathbb{R}\}$, $\delta > 0$, improves matters: by Cauchy's theorem on rectangles γ_R with $f(z) = e^{i(z^3/3+sz)}$, which is $O(e^{-\delta x^2})$ on L_δ and $O(e^{-yR^2})$ on the vertical sides,

$$\mathrm{Ai}(s) = \frac{1}{2\pi} \int_{L_\delta} e^{i(z^3/3+sz)} \, dz,$$

and since $f(z) = O(e^{-\delta x^2})$ on L_δ for every complex s, Ai extends to an entire function. It satisfies

$$\mathrm{Ai}''(s) = s\,\mathrm{Ai}(s)\,: \tag{17}$$

$\mathrm{Ai}''(s) - s\,\mathrm{Ai}(s) = \frac{1}{2\pi}\int_{L_\delta}(-z^2 - s)e^{i(z^3/3+sz)}dz = \frac{i}{2\pi}\int_{L_\delta}\frac{d}{dz}(f(z))\,dz = 0$, since f vanishes at infinity along L_δ. Comparing (17) with $y'' = Ay$ (A large) suggests $\mathrm{Ai}(s)$ decreases rapidly as $s \to \infty$ (it stays bounded by (16), excluding $e^{\sqrt{A}s}$) and oscillates as $s \to -\infty$.

Theorem 3.1

Suppose $u > 0$. Then as $u \to \infty$, (i) $\mathrm{Ai}(-u) = \pi^{-1/2}u^{-1/4}\cos\left(\frac{2}{3}u^{3/2} - \frac{\pi}{4}\right)(1 + O(1/u^{3/4}))$.
(ii) $\mathrm{Ai}(u) = \dfrac{1}{2\pi^{1/2}}u^{-1/4}e^{-\frac{2}{3}u^{3/2}}(1 + O(1/u^{3/4}))$.

Proof. (i) Change variables $x \mapsto u^{1/2}x$ with $s = -u$: $\mathrm{Ai}(-u) = u^{1/2}I_-(u^{3/2})$, where

$$I_-(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{it(x^3/3-x)}\,dx = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-s\Phi(x)}\,dx \tag{19}$$

with $s = -it$, $\Phi(x) = x^3/3 - x$. $\Phi'(x) = x^2 - 1$: two critical points $x_0 = \pm 1$; $\Phi''(x) = 2x$, $\Phi(\pm 1) = \mp 2/3$. Split into $[-2, 0]$, $[0, 2]$ (one critical point each) and $(-\infty, -2]$, $[2, \infty)$. On $[0, 2]$, Proposition 2.2 with $s = -it$, $x_0 = 1$, $\psi = 1/2\pi$, $\Phi(1) = -2/3$, $\Phi''(1) = 2$ gives $\frac{1}{2\sqrt{\pi}}e^{-i\frac{2}{3}t}\left[\frac{1}{(-it)^{1/2}} + O(\frac{1}{|t|})\right]$; the interval $[-2, 0]$ gives $\frac{1}{2\sqrt{\pi}}e^{i\frac{2}{3}t}\left[\frac{1}{(it)^{1/2}} + O(\frac{1}{|t|})\right]$ (there Φ has a maximum; apply the proposition to $-\Phi$ with $s = it$). The complementary integrals are $O(1/|t|)$ by integration by parts ($\Phi' \neq 0$ there), e.g. $\int_{-\infty}^{-2} e^{it\Phi}dx = \lim_N \frac{1}{it}\int_{-N}^{-2}\frac{d}{dx}(e^{it\Phi})\frac{dx}{\Phi'(x)}$. Adding and inserting into $\mathrm{Ai}(-u) = u^{1/2}I_-(u^{3/2})$ gives (i).

(ii) With $s = u$, $\mathrm{Ai}(u) = u^{1/2}I_+(u^{3/2})$,

$$I_+(s) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-sF(x)}\,dx, \qquad F(x) = -i(x^3/3 + x). \tag{20}$$

There is no real critical point; repeated integration by parts shows rapid decay, but not its exact order. Follow the Bessel-function principle: deform the line to a contour γ on which (a) $\mathrm{Im}(F) = 0$, and (b) $\mathrm{Re}(F)$ has a non-degenerate minimum at some z_0 (second derivative along γ positive); then $F'(z_0) = 0$. Through such z_0 pass two orthogonal curves γ_1, γ_2 on which F is real, $\mathrm{Re}(F)|_{\gamma_1}$ having a minimum and $\mathrm{Re}(F)|_{\gamma_2}$ a maximum at z_0; take $\gamma = \gamma_1$—the **method of steepest descent** ($-\mathrm{Re}\,F$ has a saddle at z_0). Here $\mathrm{Re}(F) = x^2y - y^3/3 + y$, $\mathrm{Im}(F) = -x^3/3 + xy^2 - x$, $F'(z) = -i(z^2 + 1)$, critical points $z_0 = \pm i$. At $z_0 = i$ the curves with $\mathrm{Im}(F) = 0$ are $\gamma_1 = \{y^2 = x^2/3 + 1\}$ and $\gamma_2 = \{x = 0\}$; on γ_2, $\mathrm{Re}(F)$ has a maximum at i, so take $\gamma = \gamma_1$, the branch $y = (x^2/3 + 1)^{1/2}$ of a hyperbola, asymptotic to the rays $re^{i\pi/6}$, $re^{i5\pi/6}$. Then

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-sF(x)}\,dx = \frac{1}{2\pi}\int_{\gamma} e^{-sF(z)}\,dz, \tag{21}$$

by Cauchy's theorem on the region between $\mathbb{R}$ and γ cut off by the circle of radius R (there $e^{-sF(z)} = O(e^{-cyx^2})$, so the arcs contribute $O(\int_0^{\pi/6} e^{-cR^2\sin\theta}R\,d\theta) = O(1/R)$). On γ, $\Phi(x) = \mathrm{Re}(F) = y(x^2 - y^2/3 + 1) = (\frac{8}{9}x^2 + \frac{2}{3})(x^2/3 + 1)^{1/2}$, and $dz = dx + i\frac{x}{3}(x^2/3 + 1)^{-1/2}dx$; since Φ is even and $x(x^2/3 + 1)^{-1/2}$ odd,

$$\frac{1}{2\pi}\int_{\gamma} e^{-sF(z)}\,dz = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-s\Phi(x)}\,dx. \tag{22}$$

As $(1 + u)^{1/2} = 1 + u/2 + O(u^2)$, $\Phi(x) = \frac{2}{3} + x^2 + O(x^4)$, so $\Phi(0) = 2/3$, $\Phi''(0) = 2$. Proposition 2.1 on $[-c, c]$ with $\psi = 1/2\pi$ gives $e^{-\frac{2}{3}s}\frac{1}{2\pi^{1/2}}\left[\frac{1}{s^{1/2}} + O(\frac{1}{s})\right]$; the rest is $O(e^{-2s/3}e^{-\delta s})$, since there $\Phi(x) \geq \frac{2}{3} + c_1x^2$. So $I_+(s) = e^{-\frac{2}{3}s}\frac{1}{2\pi^{1/2}}\left[\frac{1}{s^{1/2}} + O(\frac{1}{s})\right]$, which gives (ii). ∎

4 The partition function

Motivation. How many ways can n be written as a sum of positive integers? The generating function of $p(n)$ is a product of geometric series, and the saddle-point methods of this appendix give its asymptotic size.

Theorem 4.1 (Hardy–Ramanujan)

If p denotes the partition function, then (i) $p(n) \sim \dfrac{1}{4\sqrt{3}\,n} e^{Kn^{1/2}}$ as $n \to \infty$, where $K = \pi\sqrt{2/3}$.
(ii) A much more precise assertion is that

$$p(n) = \frac{1}{2\pi\sqrt{2}} \frac{d}{dn}\Big(\frac{e^{K(n-\frac{1}{24})^{1/2}}}{(n-\frac{1}{24})^{1/2}}\Big) + O(e^{\frac{K}{2}n^{1/2}}).$$

(Since $(n-\frac{1}{24})^{1/2} - n^{1/2} = O(n^{-1/2})$, $e^{K(n-1/24)^{1/2}} \sim e^{Kn^{1/2}}$, and (ii) implies (i).)

Heuristic principle: if $\sum F_n w^n$ has the unit disc as disc of convergence, the asymptotic behavior of F_n is determined by the location and nature of the "singularities" of F on the unit circle, each contributing according to its "order". (E.g. if F is meromorphic in a larger disc with only a pole of order r at $w = 1$ on the circle, then $F_n = P(n) + O(e^{-\epsilon n})$ for a polynomial P of degree $r-1$, $\sum P(n)w^n$ being essentially the principal part.)

By Euler's formula (Theorem 2.1), with $w = e^{2\pi i z}$, $z = x + iy$, $y > 0$,

$$\sum_{n=0}^{\infty} p(n) e^{2\pi i n z} = f(z) = \prod_{n=1}^{\infty} \frac{1}{1 - e^{2\pi i n z}}, \qquad p(n) = \int_\gamma f(z) e^{-2\pi i n z}\, dz, \tag{23}$$

γ the segment from $-1/2 + i\delta$ to $1/2 + i\delta$, $\delta > 0$ to be chosen in terms of n. f is largest near $z = 0$: $|f(x+iy)| \le f(iy)$ (positive coefficients), increasing as y decreases. Fortunately f is a variant of the Dedekind eta function $\eta(z) = e^{\pi i z/12}\prod(1 - e^{2\pi i n z})$: $f(z) = e^{\pi i z/12}\eta(z)^{-1}$ (the $1/12$ explains the $1/24$ in the formula). By $\eta(-1/z) = \sqrt{z/i}\,\eta(z)$ (Proposition 1.9),

$$f(z) = \sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} f(-1/z), \tag{24}$$

and as $\mathrm{Im}(-1/z) \to \infty$, $f(-1/z) \to 1$ rapidly, since

$$f(z) = 1 + O(e^{-2\pi y}), \qquad z = x + iy,\ y \ge 1. \tag{25}$$

So $f_1(z) = \sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}}$ approximates f near 0, and by (24) $p(n) = p_1(n) + E(n)$ with

$$p_1(n) = \int_\gamma \sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} e^{-2\pi i n z}\, dz, \qquad E(n) = \int_\gamma \sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} e^{-2\pi i n z}(f(-1/z) - 1)\, dz.$$

The error term. For $z \in \gamma$, since $\mathrm{Re}(i/z) = \delta/(\delta^2 + x^2)$,

$$\Big|\sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} e^{-2\pi i n z}\Big| \le c e^{2\pi n\delta} e^{\frac{\pi}{12}\frac{\delta}{\delta^2+x^2}}. \tag{26}$$

By (25) with z replaced by $-1/z$,

$$|f(-1/z) - 1| \le c e^{-2\pi\frac{\delta}{\delta^2+x^2}} \qquad \text{if } \frac{\delta}{\delta^2+x^2} \ge 1; \tag{27}$$

and since $|f(z)| \le f(iy) \le C e^{\frac{\pi}{12y}}$ for $y \le 1$ (by (24)),

$$|f(-1/z) - 1| = O(e^{\frac{\pi}{12}\frac{\delta^2+x^2}{\delta}}) = O(e^{\frac{\pi}{48\delta}}) \qquad \text{if } \frac{\delta}{\delta^2+x^2} \le 1, \tag{28}$$

as $|x| \le 1/2$. Using (26), (27) when $\frac{\delta}{\delta^2+x^2} \ge 1$ gives $O(e^{2\pi n\delta})$ (as $2\pi > \pi/12$); using (26), (28) otherwise gives $O(e^{2\pi n\delta} e^{\frac{\pi}{48\delta}})$. Choose δ to minimize, $2\pi n\delta = \frac{\pi}{48\delta}$, i.e. $\delta = \frac{1}{4\sqrt{6}} n^{-1/2}$; then

$$E(n) = O(e^{\frac{\pi}{\sqrt{6}} n^{1/2}}) = O(e^{\frac{K}{2} n^{1/2}}).$$

The main term. Add to γ the end-segments $[-1/2, -1/2 + i\delta]$ and $[1/2 + i\delta, 1/2]$, forming Γ; on them $\sqrt{z/i}\, e^{\frac{\pi i}{12z}} = O(1)$, so they contribute $O(e^{2\pi n\delta}) = O(e^{\frac{K}{4} n^{1/2}})$, absorbed in $E(n)$. So

$$p_1(n) = \int_\Gamma \sqrt{z/i}\, e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} e^{-2\pi i n z}\, dz. \tag{29}$$

Change variables $z \mapsto \mu z$ so that the exponentials combine to $e^{Ai(\frac{1}{z} - z)}$: this requires $A = 2\pi\mu(n - \frac{1}{24})$ and $A = \frac{\pi}{12\mu}$, i.e.

$$A = \frac{\pi}{\sqrt{6}} \Big(n - \frac{1}{24}\Big)^{1/2}, \qquad \mu = \frac{1}{2\sqrt{6}} \Big(n - \frac{1}{24}\Big)^{-1/2}.$$

Then

$$p_1(n) = \mu^{3/2} \int_{\Gamma'} e^{-sF(z)} \sqrt{z/i}\, dz, \qquad F(z) = i(z - 1/z), \quad s = A, \tag{30}$$

$\Gamma' = \mu^{-1}\Gamma$ the union of $[-a_n, -a_n + i\delta']$, $[-a_n + i\delta', a_n + i\delta']$, $[a_n + i\delta', a_n]$, with $a_n = \frac{1}{2}\mu^{-1} \approx \sqrt{6}\, n^{1/2}$, $\delta' = \mu^{-1}\delta \to 1$. Steepest descent: F has the critical point $z = i$ in $\mathbb{H}$; through it, F is real on the imaginary axis (maximum, rejected) and on the unit circle (minimum at i). By Cauchy's theorem replace Γ' by Γ'': the segments $[-a_n, -1]$, $[1, a_n]$ and the upper unit semicircle from -1 to 1. On the segments F is purely imaginary, so they contribute $O(a_n^{3/2}\mu^{3/2}) = O(1)$. On the semicircle, $z = e^{i\theta}$ (θ from π to 0), $dz = ie^{i\theta} d\theta$, $i(z - 1/z) = -2\sin\theta$ and $\sqrt{z/i} = e^{i(\theta/2 - \pi/4)}$; with $\theta = \varphi + \pi/2$ the integrand becomes $-e^{2s\cos\varphi} e^{3i\varphi/2}$, and the contribution is

$$\mu^{3/2} \int_{-\pi/2}^{\pi/2} e^{2s\cos\varphi} \big(\cos(3\varphi/2) + i\sin(3\varphi/2)\big)\, d\varphi.$$

Laplace's method with $\Phi(\varphi) = -\cos\varphi$, $\varphi_0 = 0$, $\Phi(0) = -1$, $\Phi''(0) = 1$, $\psi(\varphi) = \cos(3\varphi/2) + i\sin(3\varphi/2)$, $\psi(0) = 1$, gives $\mu^{3/2} e^{2s} \frac{\sqrt{2\pi}}{(2s)^{1/2}} (1 + O(s^{-1/2}))$. Since $2s = K(n - \frac{1}{24})^{1/2}$ and $\mu = \frac{1}{2\sqrt{6}}(n - \frac{1}{24})^{-1/2}$,

$$p(n) = \frac{1}{4\sqrt{3}\, n} e^{Kn^{1/2}} (1 + O(n^{-1/4})),$$

which is (i).

The precise formula (ii). Write $p_1(n) = \frac{d}{dn} q(n) + e(n)$, where

$$q(n) = \frac{1}{2\pi} \int_\Gamma (z/i)^{-1/2} e^{\frac{\pi i}{12z}} e^{\frac{\pi i z}{12}} e^{-2\pi i n z}\, dz$$

(differentiating $e^{-2\pi i n z}$ in n produces $-2\pi i z$, and $\frac{-2\pi i z}{2\pi}(z/i)^{-1/2} = (z/i)^{1/2}$), and $e(n)$ comes from the dependence of $\Gamma = \Gamma_n$ on n; by Cauchy's theorem it is $O(e^{2\pi n\delta}) = O(e^{\frac{K}{4} n^{1/2}})$. As before,

$$q(n) = \frac{\mu^{1/2}}{2\pi} \int_{\Gamma''} e^{-sF(z)} (z/i)^{-1/2}\, dz, \tag{31}$$

the segments contributing $O(a_n^{1/2}\mu^{1/2}) = O(1)$ to $\frac{d}{dn} q(n)$. On the semicircle the integral equals

$$\frac{\mu^{1/2}}{2\pi} \int_{-\pi/2}^{\pi/2} e^{2s\cos\varphi} \big(\cos(\varphi/2) + i\sin(\varphi/2)\big)\, d\varphi = \frac{\mu^{1/2}}{2\pi} \int_{-\pi/2}^{\pi/2} e^{2s\cos\varphi} \cos(\varphi/2)\, d\varphi,$$

the odd part vanishing. With $\cos\varphi = 1 - 2\sin^2(\varphi/2)$ and $x = \sin(\varphi/2)$, $\cos(\varphi/2) d\varphi = 2dx$, this is

$$\frac{\mu^{1/2}}{\pi} e^{2s} \int_{-1/\sqrt{2}}^{1/\sqrt{2}} e^{-4sx^2}\, dx, \qquad \int_{-1/\sqrt{2}}^{1/\sqrt{2}} e^{-4sx^2}\, dx = \frac{\sqrt{\pi}}{2s^{1/2}} + O(e^{-2s}),$$

and the same holds after differentiating in s. Collecting the error terms,

$$p(n) = \frac{d}{dn}\Big(\frac{\mu^{1/2}e^{2s}}{2\sqrt{\pi}\,s^{1/2}}\Big) + O(e^{\frac{K}{2}n^{1/2}}).$$

Since $\mu^{1/2}/s^{1/2} = (2\pi)^{-1/2}(n-\frac{1}{24})^{-1/2}$ and $2s = K(n-\frac{1}{24})^{1/2}$, this is (ii).

5 Selected exercises

Motivation. Two problems from Stein–Shakarchi (numbers kept): a smooth cut-off upgrades the $O(1/|s|)$ error of Proposition 1.3 to $O(|s|^{-N})$ for all $m > -1$, and the residue theorem makes the heuristic principle of Section 4 exact for meromorphic generating functions.

Exercise A.1 (Smooth cut-offs)

Let η be an indefinitely differentiable function on $\mathbb{R}$, supported in a finite interval, with $\eta(x) = 1$ for x near 0. Show that if $m > -1$ and $N > 0$, then

$$\int_0^\infty e^{-sx}x^m\eta(x)\,dx = s^{-m-1}\Gamma(m+1) + O(|s|^{-N})$$

as $|s| \to \infty$ with $\mathrm{Re}(s) \geq 0$ (principal branch of s^{-m-1}).

Solution. Fix $\delta, b > 0$ with $\eta = 1$ on $[0,\delta]$ and $\eta = 0$ on $[b,\infty)$; we may take N to be an integer. Throughout $\mathrm{Re}(s) \geq 0$, $s \neq 0$, so $|e^{-sx}| \leq 1$ for $x \geq 0$, and $e^{-sx} = -\frac{1}{s}\frac{d}{dx}e^{-sx}$.

Step 1. (Functions vanishing near 0.) If φ is indefinitely differentiable with compact support in $(0,\infty)$, then N integrations by parts (no boundary terms) give

$$\Big|\int_0^\infty e^{-sx}\varphi(x)\,dx\Big| = \Big|s^{-N}\int_0^\infty e^{-sx}\varphi^{(N)}(x)\,dx\Big| \leq \|\varphi^{(N)}\|_{L^1}|s|^{-N}.$$

Step 2. (Case $m = 0$.) One integration by parts gives

$$\int_0^\infty e^{-sx}\eta(x)\,dx = \Big[-\frac{e^{-sx}\eta(x)}{s}\Big]_0^\infty + \frac{1}{s}\int_0^\infty e^{-sx}\eta'(x)\,dx = \frac{1}{s} + O(|s|^{-N-1})$$

by Step 1 applied to $\varphi = \eta'$, which vanishes on $[0,\delta]$ and on $[b,\infty)$. Since $\Gamma(1) = 1$ this is the claim.

Step 3. (Case $-1 < m < 0$.) By the proof of Proposition 1.3,

$$\int_0^\infty e^{-sx}x^m\,dx = \lim_{R\to\infty}\int_0^R e^{-sx}x^m\,dx = s^{-m-1}\Gamma(m+1).$$

So, with $g(x) = x^m(1-\eta(x))$, it suffices to show

$$J(s) = \lim_{R\to\infty}\int_0^R e^{-sx}g(x)\,dx = O(|s|^{-N}).$$

Now g is indefinitely differentiable on $[0,\infty)$, vanishes on $[0,\delta]$, and equals x^m on $[b,\infty)$. Hence $g^{(j)}(R) = O(R^{m-j}) \to 0$ for all $j \geq 0$, and $g^{(j)} \in L^1(0,\infty)$ for $j \geq 1$ since $m - j < -1$. One integration by parts gives

$$\int_0^R e^{-sx}g\,dx = -\frac{e^{-sR}g(R)}{s} + \frac{1}{s}\int_0^R e^{-sx}g'\,dx \longrightarrow \frac{1}{s}\int_0^\infty e^{-sx}g'(x)\,dx \quad (R\to\infty).$$

Repeating with $g', \dots, g^{(N-1)}$ (the boundary terms at R tend to 0, those at 0 vanish), $J(s) = s^{-N}\int_0^\infty e^{-sx}g^{(N)}(x)\,dx$, so $|J(s)| \leq \|g^{(N)}\|_{L^1}|s|^{-N}$.

Step 4. (General $m > -1$.) Choose the integer $k \geq 0$ with $k-1 < m \leq k$ and put $\mu = m-k \in (-1, 0]$. For $0 \leq j \leq k-1$, $(x^m)^{(j)}$ is a multiple of x^{m-j} with $m-j > 0$, so it vanishes at 0; hence k integrations by parts (support of η bounded) give

$$\int_0^\infty (x^m)^{(k)} e^{-sx}\eta \, dx = (-1)^k \int_0^\infty x^m \big(e^{-sx}\eta\big)^{(k)} dx.$$

By Leibniz, $(e^{-sx}\eta)^{(k)} = (-s)^k e^{-sx}\eta + \sum_{j=1}^{k} \binom{k}{j} (-s)^{k-j} e^{-sx}\eta^{(j)}$. For $j \geq 1$, $x^m \eta^{(j)}$ has compact support in $(0, \infty)$, so by Step 1 (with $N+k$ in place of N) its contribution is $O(|s|^{k-j}|s|^{-N-k}) = O(|s|^{-N})$. With $c = m(m-1)\cdots(m-k+1)$ we have $(x^m)^{(k)} = c\, x^\mu$, and Steps 2–3 give

$$s^k \int_0^\infty e^{-sx} x^m \eta \, dx = c\, s^{-\mu-1}\Gamma(\mu+1) + O(|s|^{-N}).$$

Step 5. (Conclusion.) Divide by s^k: since $s^{-\mu-1}s^{-k} = s^{-m-1}$ for the principal branch and $c\,\Gamma(\mu+1) = \Gamma(m+1)$ by k applications of $\Gamma(z+1) = z\Gamma(z)$,

$$\int_0^\infty e^{-sx} x^m \eta(x) \, dx = s^{-m-1}\Gamma(m+1) + O(|s|^{-N-k}),$$

which is stronger than claimed. ∎

Exercise A.4 (Poles on the unit circle)

Suppose $F(w) = \sum_{n \geq 0} F_n w^n$ is meromorphic in a region containing the closed unit disc, and its only poles there lie on the unit circle, at $\alpha_1, \ldots, \alpha_k$, of orders $r_1, \ldots, r_k$. Let

$$P_j(n) = \frac{1}{(r_j-1)!}\Big(\frac{d}{dw}\Big)^{r_j-1}\big[(w-\alpha_j)^{r_j} w^{-n-1} F(w)\big]_{w=\alpha_j}.$$

(a) Show that for some $\epsilon > 0$, $F_n = -\sum_{j=1}^{k} P_j(n) + O(e^{-\epsilon n})$ as $n \to \infty$.
(b) Show that $P_j(n) = \alpha_j^{-n} Q_j(n)$ with Q_j a polynomial of degree $r_j - 1$.
(c) Find F_n exactly for $F(w) = \frac{1}{(1-w)(1-w^2)}$, and check it against the number of solutions of $a + 2b = n$ in integers $a, b \geq 0$.

Solution.

(a)

Step 1. (A larger circle.) Let Ω be the region. The closed unit disc is a compact subset of Ω, so $\{|w| \leq R_0\} \subset \Omega$ for some $R_0 > 1$. The poles of F are discrete, so only finitely many lie in $\{1 < |w| \leq R_0\}$; choose $R \in (1, R_0)$ smaller than all their moduli. Then F is holomorphic on $\{|w| \leq R\}$ except at $\alpha_1, \ldots, \alpha_k$.

Step 2. (Residues.) Put $G(w) = w^{-n-1}F(w)$. At 0, F is holomorphic with Taylor coefficients F_n, so $\operatorname{res}_0 G = F_n$. At $\alpha_j \neq 0$, w^{-n-1} is holomorphic, so G has a pole of order at most r_j; $(w-\alpha_j)^{r_j} G$ is holomorphic near α_j, and its Taylor coefficient of order $r_j - 1$ is the coefficient of $(w-\alpha_j)^{-1}$ in G. Hence $\operatorname{res}_{\alpha_j} G = P_j(n)$. The residue theorem on $|w| = R$ gives

$$F_n + \sum_{j=1}^{k} P_j(n) = \frac{1}{2\pi i} \oint_{|w|=R} w^{-n-1} F(w) \, dw.$$

Step 3. (Estimate.) With $M = \max_{|w|=R} |F|$, the right side is at most $R \cdot R^{-n-1} M = M R^{-n} = M e^{-\epsilon n}$ with $\epsilon = \log R > 0$.

(The sign matters: for $F = 1/(1-w)$ one has $F_n = 1$ and $P_1(n) = -1$.)

(b) Let $h(w) = (w-\alpha_j)^{r_j}F(w)$, holomorphic near α_j with $h(\alpha_j) \neq 0$ because the pole has order exactly r_j. Since $\left(\frac{d}{dw}\right)^i w^{-n-1} = (-1)^i(n+1)\cdots(n+i)\, w^{-n-1-i}$, Leibniz's rule gives

$$P_j(n) = \frac{\alpha_j^{-n}}{(r_j-1)!}\sum_{i=0}^{r_j-1}\binom{r_j-1}{i}(-1)^i(n+1)\cdots(n+i)\,\alpha_j^{-1-i}h^{(r_j-1-i)}(\alpha_j).$$

The sum is a polynomial $Q_j(n)$; only $i = r_j-1$ contributes to degree r_j-1, with leading coefficient $(-1)^{r_j-1}\alpha_j^{-r_j}h(\alpha_j)/(r_j-1)! \neq 0$. So $P_j(n) = \alpha_j^{-n}Q_j(n)$, and since $|\alpha_j| = 1$, $|P_j(n)|$ grows like n^{r_j-1}.

(c)

Step 1. $F(w) = \frac{1}{(1-w)^2(1+w)}$ is rational with poles 1 (order 2) and -1 (order 1). Here $w^{-n-1}F(w) = O(R^{-n-4})$ on $|w| = R$, so the integral in (a) tends to 0 as $R \to \infty$, and $F_n = -P_1(n) - P_2(n)$ exactly.

Step 2. At $\alpha = -1$: $P_2(n) = \left[\frac{w^{-n-1}}{(1-w)^2}\right]_{w=-1} = \frac{(-1)^{n+1}}{4}$.

Step 3. At $\alpha = 1$: $(w-1)^2w^{-n-1}F(w) = \frac{w^{-n-1}}{1+w}$, so

$$P_1(n) = \Big[\frac{-(n+1)w^{-n-2}}{1+w} - \frac{w^{-n-1}}{(1+w)^2}\Big]_{w=1} = -\frac{n+1}{2} - \frac{1}{4} = -\frac{2n+3}{4}.$$

Step 4. Hence $F_n = \frac{2n+3}{4} + \frac{(-1)^n}{4}$, i.e. $F_n = \frac{n}{2}+1$ for even n and $\frac{n+1}{2}$ for odd n; in both cases $F_n = \lfloor n/2\rfloor + 1$. On the other hand $F(w) = \sum_{a\geq 0} w^a \sum_{b\geq 0} w^{2b}$, so F_n counts the pairs (a,b) with $a+2b = n$, i.e. $b = 0, 1, \ldots, \lfloor n/2\rfloor$: again $\lfloor n/2\rfloor + 1$. ■